Classical Field Theory — Exercise Sheet 25

Analytic extension and direct scattering

Hints and answers

Use $L = -\partial_x^2 + u$, the Wronskian $W(f,g) = fg' - f'g$, and $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. For real $k \neq 0$, the right reflection amplitude is $r = b/a$. All potentials in this sheet are real Schwartz functions.

Exercise 25.1 (An upper-half-plane Volterra construction). For $\operatorname{Im} k \geq 0$ define

$$H(s,k) = \int_0^s e^{2ik\sigma} d\sigma, \qquad s \geq 0.$$

Thus $H(s,k) = (e^{2iks} - 1)/(2ik)$ for $k \neq 0$ and $H(s,0) = s$. Consider

$$m(x,k) = 1 + \int_x^\infty H(y-x,k)u(y)m(y,k)dy.$$

(a) For every finite R, construct the unique bounded solution on $x \geq R$ by iteration, uniformly for k in the closed upper half-plane. Prove that $m(x,k)$ is holomorphic in $\operatorname{Im} k > 0$ for each fixed x, and continuous up to its real boundary, including $k = 0$.

(b) Prove $m \to 1$ and $m_x \to 0$ as $x \to +\infty$. Show that $f_+(x,k) = e^{ikx}m(x,k)$ solves $Lf_+ = k^2 f_+$. State the two right asymptotic conditions at zero energy. Does this prove a nonzero Wronskian basis at $k = 0$?

The complex-analysis inputs may be used as givens: dominated parameter integrals of holomorphic functions are holomorphic, and locally uniform limits of holomorphic functions are holomorphic. These are the parameter-integral theorem and the uniform-limit result proved in *Analysis*, Lecture 10.

Hint. For $\operatorname{Im} k \geq 0$, $|H(s,k)| \leq s$. On $x \geq R$, bound each factor $y - x$ by $y - R$ and use ordered-simplex integration. The weighted tail $\int_R^\infty (y-R)|u(y)|dy$ is finite.

Answer. With $W_R = \int_R^\infty (y-R)|u(y)|dy$, the jth iterate is bounded by $W_R^j/j!$. The solution is holomorphic in the open upper half-plane and continuous on its closure. At $k = 0$, $f_+ \to 1$ and $f_+' \to 0$. The Wronskian formula $2ik$ gives no basis at zero.

Exercise 25.2 (Reflectionless scattering by intertwining). Fix $\kappa > 0$ and set

$$L_\kappa = -\partial_x^2 - 2\kappa^2 \operatorname{sech}^2(\kappa x), \qquad A = \partial_x + \kappa \tanh(\kappa x), \quad A^\dagger = -\partial_x + \kappa \tanh(\kappa x).$$

(a) Factor $L_\kappa + \kappa^2$ and the free operator plus κ^2. Use the resulting intertwining identity to construct both real-energy Jost solutions and determine a, b, r with the fixed Wronskian convention.

(b) Find the right-normalised negative-energy state and its inverse squared norm. Show that there are no other negative-energy L^2 states. Then give a bounded zero-energy solution which is not in L^2.

Use the Sobolev integration-by-parts and domain arguments of Lecture 18. A bounded non-L^2 zero-energy solution is called a *zero-energy resonance* in this problem.

Hint. Apply $A^\dagger$ to a free exponential and divide by its right limiting coefficient. The free negative-energy equation has no nonzero L^2 solution, so applying A to any putative bound state is restrictive.

Answer. For real $k \neq 0$,

$$f_+ = e^{ikx}\frac{k + i\kappa\tanh(\kappa x)}{k + i\kappa},$$
$$f_- = e^{-ikx}\frac{k - i\kappa\tanh(\kappa x)}{k + i\kappa},$$
$$a = \frac{k - i\kappa}{k + i\kappa}, \qquad b = r = 0.$$

The unique negative energy is $-\kappa^2$, $f_+(x, i\kappa) = \frac{1}{2}\operatorname{sech}(\kappa x)$ and its norming constant is 2κ. The zero resonance is $\tanh(\kappa x)$.

Exercise 25.3 (A weak asymmetric potential retains a phase). Let $u_\epsilon = \epsilon V$, with $\epsilon \in \mathbb{R}$, and

$$V(x) = e^{-(x-1)^2} + 2e^{-(x+2)^2}.$$

For each fixed real $k \neq 0$, derive the first-order expansions of a, b, r as $\epsilon \to 0$ and prove an $O(\epsilon^2)$ remainder bound. Evaluate the Gaussian coefficient and the leading reflection probability. Show that no negative-energy bound states occur when $\epsilon \geq 0$. The expansion is a fixed-k assertion; no uniform limit at $k = 0$ is requested.

Hint. Use the left Volterra equation and separate its sine into two exponentials. The uniform estimate for $f_- - e^{-ikx}$ is exponential in $|\epsilon|\|V\|_1/|k|$ minus one. Retain the complex Fourier integral before taking its absolute square.

Answer. The expansions are

$$a = 1 - \frac{3\epsilon\sqrt{\pi}}{2ik} + O(\epsilon^2),$$
$$b = \epsilon b_1 + O(\epsilon^2), \quad r = \epsilon b_1 + O(\epsilon^2),$$
$$b_1 = \frac{\sqrt{\pi}e^{-k^2}}{2ik}(e^{-2ik} + 2e^{4ik}).$$

The probability is $\epsilon^2\pi e^{-2k^2}(5 + 4\cos 6k)/(4k^2) + O(|\epsilon|^3)$.